

Congener-specific levels of dioxins and dibenzofurans in U.S. food and estimated daily dioxin toxic equivalent intake.



Food, especially meat, milk, and fish, is the immediate source of almost all polychlorinated dibenzo-p-dioxins (PCDDs), dibenzofurans (PCDFs), and dioxinlike compounds in the general population. To estimate intake of these highly toxic compounds, we performed congener-specific dioxin analyses for the first time on U.S. food for 18 dairy meat, and fish samples from a supermarket in upstate New York. 2,3,7,8 Tetrachlorodibenzo-p-dioxin (TCDD, "dioxin") toxic equivalents (TEqs) on a wet weight basis for the dairy products ranged for 0.04 to 0.7 ppt, meat TEqs ranged from 0.03 to 1.5 ppt, and fish TEqs ranged from 0.02 to 0.13 ppt. Previous human breast milk and infant formula analyses were used with the current preliminary food data to estimate a range of dioxin intake for Americans. Average daily food intake of TEqs for an adult weighing 65 kg was estimated to be between 0.3 and 3.0 pg/kg body weight, for a total of 18-192 pg TEq, using 1986 American consumption rates. Due to the relatively high level of PCDDs and PCDFs commonly found in human breast milk from American women and from women in other industrial countries, a nursing infant may consume an average of 35-53 pg TEq/kg body weight/day in its first year of life. This may be compared with the current U.S. EPA virtually safe dose of 0.006 pg TCDD/kg body weight per day over a 70-year lifetime based on an upper limit cancer risk of 10^{-6} , or the 10 pg/kg/day used by some European government agencies.